

RISE: Week 8 Notes

Charu Mittal

RISE: Core Intuition and Algorithm

- **The Core Problem:** Deep neural networks achieve remarkable performance, but are effectively black boxes.
- We discussed a motivating example where the model only activates on white sheep for the “sheep” class, and confuses black sheep with a cow, likely due to training data scarcity.
- To explain CNNs, we generally use **saliency maps**. There are two broad families:
 1. **White-box:** Uses internal model information, gradients, activations, weights (e.g., Grad-CAM, CAM).
 2. **Black-box:** Uses only input/output access. More general and model-agnostic.
- **RISE** is a black-box approach.
- A limitation of white-box methods is that they are architecture-specific. If the architecture changes, the method must be reimplemented.
- Black-box methods do not require internal model details (weights or parameters); they only need predictions.
- **Key Idea:** If a pixel is important, hiding it should reduce the model’s confidence. If it is irrelevant, masking it should have little effect.
- **RISE Approach:**
 1. Generate N random binary masks $\{M_i\}$
 2. Multiply each mask element-wise with the image: $I \odot M_i$
 3. Query the black-box model to obtain scores: $f(I \odot M_i)$
 4. Aggregate importance as a weighted sum of masks, where weights are the output scores
- RISE suggests using **bilinear upsampling** due to problems with pixel-independent mask because:
 1. Mask space is too large: $2^{H \times W}$ (exponential)
 2. Sharp binary pixel patterns can cause adversarial artifacts
 3. A single pixel flip can drastically change the DNN output
- **RISE Solution: Upsampling Smaller Masks**
 1. Sample a binary mask of size $h \times w$ where $h \times w \ll H \times W$, with probability $P(\text{cell} = 1) = p$
 2. Upsample to $(h + 1)C_H \times (w + 1)C_W$ using bilinear interpolation
 3. Apply a random shift in $(0, C_H)$ and $(0, C_W)$
 4. Crop to size $H \times W$

Mathematical Formulation

definition: The importance of pixel is the expected model output over all masks, conditioned on pixel being visible:

$$S_{I,f}(\lambda) = \mathbb{E}_M [f(I \odot M) \mid M(\lambda) = 1] = \sum_m f(I \odot m) P[M = m \mid M(\lambda) = 1] \quad (1)$$

that can also be derived in this form :

$$S_{I,f}(\lambda) \stackrel{\text{MC}}{\approx} \underbrace{\frac{1}{\mathbb{E}[M] \cdot N}}_{\text{normalisation}} \sum_{i=1}^N \underbrace{f(I \odot M_i)}_{\text{confidence weight}} \cdot \underbrace{M_i(\lambda)}_{\text{pixel visibility}} \quad (2)$$

Where:

- $\frac{1}{\mathbb{E}[M]}$ is the normalisation term
- $f(I \odot M_i)$ is the confidence weight
- $M_i(\lambda)$ represents pixel visibility

Basically this tells, When pixel λ is unmasked ($M(\lambda) = 1$), how confident is the model on average.

So it gives: **High importance** $\Rightarrow$ model is consistently more confident when this pixel is visible.

Low importance $\Rightarrow$ visible or not, the score barely changes.

Connection to LIME

- **LIME** (Ribeiro et al., 2016) is also a black-box method. It fits a local linear model g near the input by sampling perturbations and minimising:

$$\min_{g \in G} \sum_{z \in Z} \pi_x(z) (f(z) - g(z))^2 + \Omega(g)$$

Importance is given by the coefficients of g . LIME operates at the **superpixel level**, not pixel level.

- **RISE vs. LIME comparison:**

Aspect	RISE	LIME
Granularity	Pixel-level	Superpixel-level
Model	Non-parametric	Linear approximation
Masks	Smooth random	Segment-based
Bias	None	Linear assumption

Evaluation Metrics: Deletion and Insertion

- RISE proposes its own **causal metrics** that evaluate explanations by intervening directly on the image. Two metrics are defined:

- **Deletion Metric**

1. Order pixels by *decreasing* importance according to saliency map S
2. Progressively set them to a constant grey value (not blur, to ensure the pixel is truly removed)
3. Record the model confidence $f(\text{modified image})$ at each step
4. Score = Area Under the Curve (AUC) — *lower is better*

Intuition: If truly important pixels are removed first, model confidence should drop sharply. Constant grey (not blur) is used because blur still leaves low-frequency signal.

- **Insertion Metric**

1. Start with a *highly blurred* version of the image (not a blank canvas, to avoid shape-edge artifacts)
2. Order pixels by *decreasing* importance from S
3. Progressively reveal them from the original image
4. Record model confidence at each step
5. Score = AUC — *higher is better*

Intuition: If truly important pixels are revealed first, model confidence should rise quickly.

Algorithm 1

```

1: procedure DELETION
2:   Input: black box  $f$ , image  $I$ , importance map  $S$ , number of pixels  $N$  removed per step
3:   Output: deletion score  $d$ 
4:    $n \leftarrow 0$ 
5:    $h_n \leftarrow f(I)$ 
6:   while  $I$  has non-zero pixels do
7:     According to  $S$ , set next  $N$  pixels in  $I$  to 0
8:      $n \leftarrow n + 1$ 
9:      $h_n \leftarrow f(I)$ 
10:   $d \leftarrow \text{AreaUnderCurve}(h_i \text{ vs. } i/n, \forall i = 0, \dots, n)$ 
11:  return  $d$ 

```

(a) Deletion Algorithm

Algorithm 2

```

1: procedure INSERTION
2:   Input: black box  $f$ , image  $I$ , importance map  $S$ , number of pixels  $N$  removed per step
3:   Output: insertion score  $d$ 
4:    $n \leftarrow 0$ 
5:    $I' \leftarrow \text{Blur}(I)$ 
6:    $h_n \leftarrow f(I)$ 
7:   while  $I \neq I'$  do
8:     According to  $S$ , set next  $N$  pixels in  $I'$  to corresponding pixels in  $I$ 
9:      $n \leftarrow n + 1$ 
10:     $h_n \leftarrow f(I')$ 
11:   $d \leftarrow \text{AreaUnderCurve}(h_i \text{ vs. } i/n, \forall i = 0, \dots, n)$ 
12:  return  $d$ 

```

(b) Insertion Algorithm

Figure 1

Extension: RISE for Image Captioning

- RISE naturally generalises to **sequence models** such as image captioning. Given a captioning model $f(I, s, w_k) = P[w_k \mid I, w_1, \dots, w_{k-1}]$, RISE computes a per-word saliency map:

$$S_{w_k}(\lambda) = \frac{1}{N \cdot \mathbb{E}[M]} \sum_{i=1}^N f(I \odot M_i, s, w_k) \cdot M_i(\lambda) \quad (3)$$

- This produces a separate saliency map for *each word* in the caption.
- RISE is not constrained to fixed-size patches and better handles varying object sizes.
- **Example:** For the caption “A horse and carriage on a city street”, RISE correctly highlights the horse region for the word “horse” and the carriage region for the word “carriage”.

Failure Cases and Limitations

- **Background noise:** MMonte Carlo approximation with finite N leaves residual noise, especially around object boundaries.
- **Small objects:** Hard to isolate with 7×7 coarse masks, masks; importance can leak to surroundings.
- **High computational cost:** $N = 4000\text{--}8000$ forward passes per image.
- **Multi-object scenes:** Importance maps can be diffuse when multiple relevant objects overlap.

Summary

RISE in one sentence: Probe the model with randomly masked images, then reconstruct pixel importance as a weighted average of the masks, weighted by the model’s output on each masked image.

1. **RISE algorithm:** Black-box pixel saliency via Monte Carlo masking, no gradients, no internals, no architecture assumptions.
2. **Smooth mask generation:** Bilinear upsampling + random shift avoids adversarial artefacts and reduces mask-space explosion from $2^{H \times W}$.
3. **Causal evaluation metrics:** Deletion and Insertion are human agnostic metrics.
4. **State-of-the-art results:** Outperforms all black-box methods; competitive with white-box methods.
5. **Captioning extension:** Naturally generalises to sequence models.